MIL-C-48881B (AR)
11 March 1983
SUPERSEDING
MIL-C-48881A
30 October 1978

MILITARY SPECIFICATION

CHARGE, PROPELLING, M205
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions for the loading, assembling and packing for the Charge, Propelling, M205 for use on the Cartridge, 81MM, HE, M374A3 and WP, M375A3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9278773	- Container Assembly
9280588	- Charge, Propelling, M205
9313721	- Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M205 or Container Assembly
9313756	- Packing for Pallet for Box, Fiberboard, Outer for Charge Propelling M205 or Container Assembly

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The propelling charge assembly shall comply with all requirements specified on drawing 9280588 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Gross weight of propelling charge. The weight of the loaded propelling charge shall not vary more than plus or minus

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eleven grains from the assessed propellant weight plus the median drawing weight of the container assembly when tested as specified in 4.5.3.

3.4 Velocity. The propelling charges shall impart a corrected muzzle velocity of not more than 900 feet per second (fps) and not less than 870 fps when tested as specified in 4.5.4 (see 6.9).

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.

3.6.1 Propelling charge. The propelling charge shall be free of dirt, grease or other foreign material.

3.6.2 Packaging. The packaging shall be free from foreign material. The boxes shall be assembled without undue binding and no flap shall project beyond the edge of the box when closed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following item in sample quantities as indicated.

Part Description	Drawing	Quantity
Charge, Propelling, M205	9280588	210

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 (AR).

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Charge, Propelling, M205				9280588
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	110		3.2	4.4.2.1
	Weight of propellant	10		3.2	4.5.2
	Gross weight of propelling charge	110		3.3	4.5.3
	Velocity (Proving Ground)	*200		3.4	4.4.3.1/4.5.4

NOTES:

*200 is equivalent to the max. charge of 50 cartridges (4 charges per cartridge).

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of propelling charges shall contain:

a. Container parts of one lot interfix from one manufacturer.

b. Propellant from not more than one lot.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.1	Charge, Propelling, M205				9280588
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Weight of propellant in excess of 10 grains below min.	(a)	100%	3.2	4.5.1
2	Gross weight of propelling charge, min.	(b)	100%	3.3	4.5.3
3	Container or container seal damaged to the extent that propellant shall escape		100%	3.2	Visual(4.5.5)
Major					
101	Weight of propellant from min. to and including 10 grains below min.	(a)	100%	3.2	4.5.1
102	Weight of propellant, max.	(a)	100%	3.2	4.5.1
103	Gross weight of propelling charge, max.	(b)	100%	3.3	4.5.3
104	Velocity (Proving Ground)	200/100(c)	1.0 %	3.4	4.4.3.1/4.5.4
105	Container or container seal damaged but not to the extent that propellant shall escape		100%	3.2	Visual(4.5.5)
Minor					
201	Evidence of poor workmanship		0.65%	3.6.1	Visual

NOTES: (a) This inspection may be performed prior to loading. See 4.5.1 for selection of test method. (b) In the automated mode, this weight is the total loaded container weight minus the empty container weight for each loaded increment. (c) 200/100 is equivalent to the max. charge of 50/25 cartridges (4 charges per cartridge).

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.2	Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M205 or Container Assy.		9313721
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charges incorrectly packed		0.40%	5.1	Visual
102	Barrier bag damaged (punctured)		0.40%	5.1	Visual
103	Marking improper (barrier bag, outer box)		0.40%	5.1	Visual
104	Improper seal (barrier bag, inner and outer boxes)		0.40%	5.1	Visual
105	Orientation stripe on corner of outside box improperly located (with respect to loading holes of charges)		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.3	Packing for Pallet				9313756
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of boxes improper		0.40%	5.2	Visual
102	Strapping loose		0.40%	5.2	Visual
103	Boxes damaged		0.40%	5.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing.

4.4.3.1 Velocity.

4.4.3.1.1 First article. The sampling plan for cartridges required and procedure/acceptance criteria for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 0.40 percent. The test shall be performed as specified in 4.5.4.

4.4.3.1.2 Production.

4.4.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots of propelling charges have complied with the applicable requirements, the sampling plan and required procedures for cartridges for this test shall be in accordance with MIL-STD-414 Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4.

4.4.3.1.2.2 After three consecutive lots have met the criteria of 4.4.3.1.2.1 the sampling plan for cartridges for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I, and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4. In the event of a lot rejection the sampling plan of 4.4.3.1.2.1 shall be required beginning with the next lot and continuing until three consecutive lots of propelling charges have complied with the applicable requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test method and procedures.

4.5.1 Weight of propellant.

4.5.1.1 Manual method.

4.5.1.1.1 Method 1. The propellant shall be weighed. Weight limits for weighing should be established within the applicable weight requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirement shall be classified defective and removed from the lot.

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4.5.1.1.2 Method 2. Samples for this test shall be selected from propelling charges that have met the requirements of 3.3. A sample of 200 propelling charges shall be randomly selected during production for each shift. The propellant shall be removed from the charge, and weighed. If one or more critical defects or two or more major defects are found, the production represented by the sample shall be rejected.

4.5.1.2 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. The automated equipment shall weigh the container and place this weight in memory and then weigh/check-weigh the propellant before loading. After loading, the total weight of the container/propellant combination shall be weighed to verify complete loading by subtracting the increment weight (from memory) from the total weight to obtain the loaded propellant weight. In such cases, process control plans in accordance with Data Item Description DI-P-1604 must be submitted to the Technical Agency (see 6.6) for review of the automated operation.

4.5.2 Weight of propellant in charge, propelling (first article).

4.5.2.1 Destructive test. The propellant shall be removed from the charge, propelling and weighed. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

4.5.2.2 Non-destructive (alternate procedure). The charge, propelling shall be weighed and the net weight of propellant determined by subtracting parts tare weight from the gross weight. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

4.5.3 Gross weight of propelling charge. The loaded propelling charge shall be weighed individually. If any assembly fails to meet the requirement, it shall be classified defective and removed from the lot.

4.5.4 Velocity test. This test shall be conducted at a Government proving ground. The rounds shall be inert loaded to 9.05 plus or minus .05 pounds including the weight of the dummy fuze. The rounds shall be assembled at maximum charge (i.e. with four charges). The rounds shall be conditioned to 70° $\pm$ 2.0° Fahrenheit and fired from a hard mount at an elevation of 800 mils, from an (M29) 81MM mortar tube (3.207 inches max.)

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for which the round is standard. Reference rounds shall be fired after every third test round. These reference rounds shall consist of reference standard components and a calibration lot of propellant charges, and ignition cartridges for velocity correction. The individual velocities and pressures shall be obtained and recorded. Velocity standard deviation and mean velocity shall be calculated to determine compliance with applicable requirements. A formal firing report shall be submitted (see 6.7).

4.5.5 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedures shall be used:

a. Place the damaged charge on a M24 fin, oriented such that the propellant would be resting against the damaged surface.

b. The fin-charge assembly shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed, the damaged propelling charge shall be classed as a major defect.

5. PREPARATION FOR DELIVERY

5.1 Unit packing. Packing shall be as specified on dwg. 9313721.

5.2 Exterior packing. Packing shall be as specified on dwg. 9313756.

5.3 Marking. Marking shall be as specified on dwg. 9313721 and dwg. 9313756.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 81MM Mortar for use on the Cartridge, 81MM, HE, M374A3 and WP, M375A3.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

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6.4 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.1.2	Manufacturing Methods Reports	DI-P-1604

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Submission of process control plans for approval. Submit plans as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-B, Dover, NJ 07801.

6.7 Submission of test data. Formal firing reports containing velocity test data and lot acceptability shall be sent to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-B, Dover, NJ 07801.

6.8 Proving ground test summary.

Test	Sample Size	Requirements
First Artricle		
Velocity	200 Propelling Charges (50) 81MM M374A3 Inert Cartridge and Fuzes W/ Acceptable Ignition Cartridge, and 17 81MM M374A3 Inert Cartridges and Fuzes W/Acceptable Ignition Cartridges and Calibration Lot of Propelling Charges (68 Charges)	See Paragraph 3.4

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First Three Production Lots		
Velocity	200 Propelling Charges (50) 81MM M374A3 Inert Cartridge and Fuzes W/Acceptable Ignition Cartridges, and 17 81MM M374A3 Inert Cartridges and Fuzes W/Acceptable Ignition Cartridges and Calibration Lot of Propelling Charges (68 Charges)	See Paragraph 3.4
Subsequent Production		
Velocity	100 Propelling Charges (25) 81MM M374A2 Inert Cartridges and Fuzes W/Acceptable Ignition Cartridges, and 9 81MM M374A3 Inert Cartridges and Fuzes W/Acceptable Ignition Cartridges and Calibration Lot of Propelling Charges (36 Charges)	See Paragraph 3.4

6.9 Velocity requirement. The velocity requirement in this specification is established for testing mortar cartridges fired from a hard mount (armor plate). Since all velocity assessments will be conducted from hard mounts, a correction factor applied to the assessed velocities will reflect the appropriate firing table velocities when a standard mortar base plate is used.

Custodian:
Army-AR

Preparing activity:
Army-AR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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